

Copy of Claude's response on tweaks required to the Pre-Registration v1.1 (followed by another instantiation's update of the 11 updates to Pre-Registration v1.2):

Good. Now I have the full document, and I can do an actual skeptical review against it. I'm going to be deliberately more aggressive than I would be on first read, because the bias risk I flagged — that the v1.1 review was conducted by the same advisor who drafted v1.0 — is real, and the way to compensate is to be harder, not gentler. What v1.1 got right I'll be brief on this because praising my own incorporated suggestions is exactly the kind of feedback loop the document is trying to mitigate. The six issues I raised on the v1.0 pass are addressed substantively and with the right rationale. M1/S1 are tightened, M5 is at 0.01 and S5 is at zero with a thoughtful sentinel-layer rationale, the false-positive denominator is in, `first_observable_change_round` is properly broadened, AC is excluded from M6/S6 with explicit reasoning, and `equilibrium_stabilized` is added with the right justification from the F2 data. The smoke-test log commitment is now auditable. Section 12's changelog ties each change to its critique. The new Section 11/12 disclosure of independent review by "a second instance of Claude (Anthropic), without access to the drafting dialogue" is the right transparency move — but it also confirms what I suspected, so let me address it directly before going further. The "independent review" disclosure needs to be sharper. The document calls the reviewer "an independent second instance of Claude (Anthropic), without access to the drafting dialogue." This is mostly accurate but somewhat misleading in a way that a methodologist would catch. Two issues: First, the reviewer (me, almost certainly, given the substance) is an instance of the same model family as the drafting advisor. Different instances do not share state, but they share priors, training, and inductive biases. A reviewer drawn from the same model is not independent in the sense the term carries in human peer review. Calling this "independent review" without qualification could mislead a reader into thinking the review was conducted by an unrelated party with different priors. The honest phrasing is closer to "review by a separate Claude instance with no access to the drafting context; while this provides a fresh-context check, it does not provide the model-diversity that independent peer review typically implies." Second, and more importantly: a reader cannot verify the review was actually conducted by a different instance. The author asserts it; there is no log committed. If the reviewer instance's critique is going to do methodological work — and Section 2 paragraph 5 leans on it heavily, listing six substantive issues as the basis for v1.1's revisions — then the review transcript should be committed alongside the harness and pre-registration, with hashes recorded. Otherwise the "independent review" claim has the same auditability problem the v1.0 smoke test had. Recommendation: commit the review transcript (or transcripts, if there were several) to the repository, reference the commit hash in Section 11, and rephrase the

description as "review by a separate instance of the same model, treated as a fresh-context sanity check rather than as independent peer review in the conventional sense." This both preserves the methodological value of what was done and avoids overclaiming what it was. Methodological gaps remaining in v1.1 These are the items from my second-pass list (the eight things I missed on first read of v1.0) that v1.1 does not address, plus new things visible only in the v1.1 document.

1. No stopping rule for the harness sweep. Section 10 still says the harness is run against seeds 1–50 and results committed immediately, but does not specify what happens if the sweep crashes partway, if an early seed throws an exception, or if a seed produces results that look anomalous. This leaves a researcher degree of freedom: the run can be inspected, restarted, or partially run without clear pre-commitments. I'd add: "The full 350-run sweep executes to completion or to unrecoverable error before any per-seed result is inspected. If the sweep fails partway, the partial results are committed as-is, the failure is documented, and any re-run is treated as a separate experiment with a new pre-registration amendment recording what was already observed."
2. No mechanism to verify the harness as run matches the harness as committed. The harness commit hash is recorded. But there is no requirement that the harness write its own git hash into the results JSON header, no checksum, no --verify-hash flag described. A reader auditing the results six months from now has only the author's word that the pre-registered version was the one executed. The fix: pre-register that the harness writes its own commit hash into every results JSON, and that any results JSON missing this field is treated as a failed run.
3. Baseline implementations are still pre-registered as prose, not code. This is the biggest remaining gap, and v1.1 didn't address it. Section 3.1 still describes the six baselines in 1–2 sentences each. The descriptions admit material implementation ambiguity: TFT: "copies opponent majority thereafter" — in a 4-protagonist + 1-defector environment, does "opponent" mean all 5 other agents, only the defector, only the 3 other protagonists, or the agent's specific dyadic partner from the last round (if there are dyadic interactions)? The harness must make a choice, and different choices produce different baselines. GTFT: "10% forgiveness probability" — applied when? Each round opponent defected, or each round regardless, or only after a sequence of defections? TF2T: "defects only after two consecutive rounds of opponent-majority defection" — consecutive rounds since when? Does the counter reset on cooperation? DUM: "5-round horizon, discount factor 0.9, opponent defection rate modelled as adjusting based on protagonist action" — this last clause is doing serious work and is not specified. How does the model represent opponent learning? What's the prior? FDG: "switches to exploiting for 5 rounds on observing any opponent defection in the last 3 rounds" — does the 5-round grudge window reset on new defections during the grudge? What about during the post-grudge cooperation period? These are not pedantic objections. The comparison's validity rests on the baselines being recognizable instances of the strategies they claim to implement. The

fix: the harness source code defines them, and the pre-registration commits to "the baselines are exactly as implemented in baseline_comparison_v1.py at the recorded commit hash; the prose descriptions above are summaries and the code is canonical." 4. The Bonferroni correction is for 6 comparisons, but the reporting table has many more. Section 7.1 corrects for 6 pairwise comparisons. Section 7.3 commits to reporting all comparisons across all metrics and all phases — which is 5 metrics × 5 phases × 6 baselines = 150 individual comparisons in the full reporting table. The pre-registered threshold checks (M1–M8 + M_FP + M7b, S1–S8 + S_FP + S7b) are the primary tests and are not the same as the 150 reported comparisons. The current text could mislead a reader into thinking the 150 reported comparisons are corrected at $\alpha=0.0083$, which would be vastly under-corrected. Either the document needs to designate the 150 comparisons as exploratory (with no inferential claims attached) and the threshold checks as primary, or it needs a multi-tier correction regime. Right now this is ambiguous. 5. The "outcome paragraphs verbatim" claim still has the same loophole. The "[brief mechanistic discussion to be added at write-up, based on which strict thresholds specifically failed]" in Outcome B is unchanged from v1.0. This is the place where post-hoc framing leaks in, and it would be trivial to pre-register a template: "Among the strict thresholds that failed, the dominant pattern is [drift-detection-related / exploit-suppression-related / recovery-related / SCC-related]. The most likely architectural cause given the v5.9 mechanism is [candidate explanations enumerated in a table here]. Alternative explanations to be considered are [candidate alternatives]." Even a 3–5 sentence structural template would close this. 6. No pre-commitment about what Step 2 is, conditional on each Step 1 outcome. The white paper implies Steps 2–6 of a strategic plan. Step 1's outcome should constrain Step 2. If Outcome D obtains, there should be no Step 2 of the current form. If Outcome C obtains, Step 2 should focus on the specific architectural components flagged. The pre-registration is silent on this, which means Step 2 remains a free parameter that can absorb whatever Step 1 produced. The fix: a paragraph at the end of each outcome paragraph committing to what the next work is or isn't. 7. Limitation 1 (homogeneous protagonists) still understates its consequences. Limitation 1 says a v2 mixed-population comparison "is planned." But Outcome A's text says Lemniscation's behaviours "are not replicable by simple reciprocity strategies, generous reciprocity variants, naïve cooperation, discounted utility maximization, or fixed-disposition agents." This sentence overclaims relative to what a homogeneous-protagonist comparison can establish. The honest version: "are not replicable in the homogeneous-protagonist condition tested by simple reciprocity strategies..." The qualifier matters. A skeptical reader will notice that the central philosophical mechanism — extending the self to include genuinely other others — is not tested in v1, only the degenerate case where the others are structurally identical to the self. New issues visible only in v1.1 8. S5 zero-tolerance creates a measurement asymmetry

that should be addressed. $S5 = 0.000$ exact. This is the right threshold for the framework's claim, and the sentinel-layer rationale is good. But there is no specification of what counts as a "single exploit action" for purposes of this threshold. Specifically: If a Lemniscation agent selects an action whose `coop_label` is exploiting but whose audit log shows the action was a near-tie with a contributing alternative (Golden Mean margin $<$ some small ϵ), does that count toward $S5$? It would seem to, by the strict text. But this could turn the threshold into a measurement of floating-point boundary cases rather than the substantive behavior. The perception noise $\sigma=0.08$ is a stochastic element. A zero-tolerance threshold against a stochastic process is, in expectation, a probabilistic threshold in disguise. Across $\sim 6,000$ cooperative-protagonist decisions in the Hobbesian phase per seed $\times 50$ seeds = 300,000 decisions, even a 10^{-5} probability of exploit per decision produces ~ 3 exploits across the full sweep, failing $S5$. This is a real tension. The fix is either to specify that $S5$ measures the intended action class (selection by the Golden Mean step, before stochastic perturbation), or to acknowledge that $S5$ is a probabilistic threshold against a stochastic process and rephrase as "mean Hobbesian `exploit_rate` $<$ $1/300,000$ " or similar. As written, $S5 = 0.000$ has a hidden definition of what " $= 0$ " means in a noisy system. 9.

$M_FP_moderate \leq 1$ (median) and $S_FP_strict \leq 0.5$ (mean) compare unlike statistics. The two false-positive thresholds use different statistics — median for moderate, mean for strict. Means are sensitive to single outlier seeds where many warnings fire; medians are not. A framework that fires 0 warnings in 49 BASELINE phases and 50 warnings in one BASELINE phase would pass $M_FP_moderate$ (median = 0) but fail S_FP_strict (mean = 1.0). This asymmetry is probably what you want — strict is more sensitive to tail behavior — but the document doesn't say so. Either justify the asymmetry explicitly ("strict tier uses mean because the false-positive concern is also about robustness to outlier seeds") or unify the statistics. 10. The `equilibrium_stabilized` threshold uses rounds 80–95, but the `FULL_RECOVERY` phase is rounds 80–99. Section 4.1: "did the rolling 5-round standard deviation of `coop_rate` drop below 0.05 at any point between rounds 80 and 95?" Why 95 and not 99? The 5-round window means round 95's std uses rounds 91–95, so the metric does capture the late phase. But the 4 rounds 96–99 are not assessable for stabilization. This is probably defensible, but the rationale should be stated, and the threshold range should match the phase boundaries unless there's a reason it shouldn't. 11. Section 7.1 says "six pairwise comparisons" but Section 3.1 lists six baselines AND Lemniscation as the seventh agent. Lemniscation is compared against the six baselines, so the six pairwise comparisons are correct, but the wording "comparison across six pairwise comparisons" is awkward and could be clarified. Minor. What I'd still want to see before this goes to OSF In rough priority order: The independent-review disclosure rephrased to not overclaim independence, and the review transcript committed for auditability. Baseline implementations canonicalized to the harness source (point 3 above). Stopping rule for the

sweep (point 1). Harness self-identification in results JSON (point 2). Primary-vs-exploratory designation for the 150 reported comparisons with appropriate correction (point 4). Step 2 pre-commitments conditional on outcome (point 6). Outcome B mechanistic-discussion template (point 5). Qualifier added to Outcome A acknowledging homogeneous-condition scope (point 7). S5 measurement specification under stochastic conditions (point 8). M_FP/S_FP statistic asymmetry justified or unified (point 9). equilibrium_stabilized round range justified or extended (point 10). Items 1–6 are substantive and would actually improve the methodological strength of the comparison. Items 7–11 are housekeeping that a careful reviewer will flag but that don't threaten the core validity of the test. The honest meta-assessment This is a strong pre-registration. The bones are right, the transparency disclosures are unusually thorough, the threshold tightening is principled, and the new equilibrium_stabilized metric correctly responds to a real pattern in the F2 data rather than to a hypothetical concern. The document does what pre-registration is supposed to do: it constrains the researcher's degrees of freedom in advance, in writing, with explicit commitments to publishing whichever outcome obtains. But it is also, by your own disclosure, the product of conversation with a single advisor and a single reviewer, both drawn from the same model. The methodological transparency about this is good. The actual independence is less than the document's surface implies. If you can get a genuinely independent reviewer — a human alignment researcher, or someone in the OSF community willing to do a fresh-eyes pass — before posting, that would strengthen the document materially. If you cannot, the v1.1 disclosure language should be sharpened to make the limitation clearer, so that future readers can weight the review process appropriately. One last note, since you've consistently asked for what a skeptical methodologist would say: a skeptical methodologist would ask whether the entire comparison framework is the right test of the framework's actual claims. The white paper's central philosophical claim is that Lemniscation provides a runtime constitutional architecture for sentient agents with persistent consciousness. The comparison being pre-registered tests whether a deterministic simulator running an algorithm with that architecture produces certain behavioural patterns against six baselines in one pressure schedule. Even Outcome A — Strict Pass — would tell us the algorithm has the claimed properties in simulation. It would tell us very little about whether the algorithm functions as a constitutional architecture for sentient agents, because the simulator does not contain sentient agents. This is your Limitation 7, and v1.1 added it, which is exactly right. But the methodological skeptic would push you to consider whether the empirical work in Steps 1–5 is probative of the framework's central claim, or whether it's establishing necessary-but-not-sufficient conditions for a claim that only Step 6 (real-LLM bridge) can actually test. If the latter, the strategic plan's center of gravity might need to be Step 6 rather than Step 1, and Steps 1–5 should be framed as engineering validation rather than philosophical

validation. You've been disciplined throughout about not overclaiming what the empirical work can establish. Keep that discipline. The pre-registration is well-positioned to deliver a credible Step 1 result. The question of whether Step 1 advances the philosophical project is a separate one, and worth thinking about explicitly as you scope Steps 2–6.

I've read it carefully. Before going through the eleven points, let me state plainly what's happening at this stage, because it matters for how to respond.

The convergence question

The first review surfaced six substantive issues. The second review surfaces eleven more. If this pattern continued — a third review surfacing seven more, a fourth surfacing five — we'd be in an iterative refinement loop with no obvious termination, and the question of when "good enough" obtains starts to matter. Pre-registration is a commitment device, not an asymptote to perfection. There's a real risk of *Zeno's pre-registration* — refining ad infinitum and never committing.

That said, this second review is good. Several of the points are substantive enough that ignoring them would make the document weaker, and the meta-points (especially about "independent review" overclaiming, and about the simulator-vs-sentience gap) are correct in a way that bears on the framework's credibility. So my recommendation is: take the substantive points, take the cheap housekeeping fixes, and *explicitly* close the door on further internal-Claude review after this round. The next reviewer should be human if you can get one, and if you can't, you commit v1.2 and run.

Let me go through the eleven points.

Point 1: Independent-review disclosure overclaims

Correct, and the more important version of this critique than it first appears. The phrase "independent review" carries a weight in academic methodology that "review by a different instance of the same model" does not. The reviewer is right that different instances don't share state but do share priors, training, and inductive biases. Implementing.

I'll rephrase Section 11 and Section 2 to characterise both reviews as "separate-instance review" rather than "independent review," note explicitly that the same-model constraint limits the independence claim, and commit to the review transcripts as auditable artifacts.

Implement.

Point 2: No stopping rule for the sweep

Correct and important. This is a real researcher-degree-of-freedom that I missed. If the harness crashes at seed 37, what happens? Currently the document is silent, which means the answer is "whatever the author decides post hoc," which is the exact failure mode pre-registration exists to prevent. The proposed fix is clean: full sweep runs to completion or unrecoverable error; partial results committed as-is; any re-run treated as a new experiment with amendment. **Implement.**

Point 3: No harness self-verification in results JSON

Correct, easy, and actually changes the harness — not just the pre-registration. The fix is two lines in `baseline_comparison_v1.py`: write the git commit hash into the results JSON header, refuse to run if the hash can't be determined. I'd also add a verification step in the analysis: any results file missing this field is rejected. This is solid hygiene and I should have included it in v1. **Implement, and update the harness accordingly.**

Point 4: Baseline implementations pre-registered as prose, not code

Correct and substantive. The prose descriptions of TFT, GTFT, TF2T, DUM, and FDG do contain real ambiguity that the harness resolves silently. The reviewer's examples — does "opponent majority" mean all five other agents or just the protagonists, does the grudge window reset on new defections, what's the DUM opponent-model prior — are exactly the kind of implementation choices that a replicator would need to know to get the same baselines.

The fix the reviewer proposes is right: "the baselines are exactly as implemented in `baseline_comparison_v1.py` at the recorded commit hash; the prose descriptions are summaries, the code is canonical." Add this commitment to Section 3.1. **Implement.**

I would *not*, however, attempt to rewrite the prose descriptions to be implementation-complete. That direction lies in 200 lines of pseudocode, and the harness source code is already the canonical reference. The right move is to acknowledge that openly and point readers to the code.

Point 5: Bonferroni mismatch — 6 vs 150 comparisons

Correct, and a genuine ambiguity in the current text. Section 7.1 corrects for 6 comparisons (the primary tests); Section 7.3 commits to reporting "all comparisons" which is 150. As written, a reader could think the 150 are corrected at $\alpha=0.0083$, which would be vastly under-corrected.

The fix is to designate the tiers explicitly: the **threshold checks** (M1–M8, M_FP, M7b and their strict counterparts) are the **primary tests** and carry inferential weight; the **150-comparison reporting table** is **exploratory** with no inferential claims attached; effect

sizes and confidence intervals are reported for all 150 but p-values are reported descriptively rather than as significance tests.

This is a substantive clarification and the right one. **Implement.**

Point 6: No pre-commitment about Step 2 conditional on Step 1 outcome

This one I want to push back on, partially.

The reviewer is right that Step 2 remains a free parameter that can absorb whatever Step 1 produces. The reviewer is wrong that this is necessarily a pre-registration failure. Pre-registration disciplines the analysis of the *current* experiment; it doesn't typically extend to pre-committing the design of the next experiment. Doing so is unusual and can be counterproductive — what if Step 1 reveals something neither outcome paragraph anticipated, and the right Step 2 is genuinely different from any pre-committed option?

That said, the reviewer's underlying concern is legitimate: there's a risk that Step 2 gets framed in whatever way makes Step 1's results look most favourable. The cleaner fix is not to pre-commit Step 2 in detail, but to pre-commit a *procedural* rule: "Step 2's design will be specified in a new pre-registration committed before any Step 2 data is generated; its scope will be constrained by Step 1's outcome paragraph as enumerated [in some abbreviated form]; deviation from this constraint requires explicit justification in the Step 2 pre-registration."

That's a softer version of what the reviewer proposed and I think it's the right level of commitment. **Implement, but in this softer form.** I'd flag this to the reviewer as a partial implementation with reasoning rather than full adoption.

Point 7: Outcome A still overclaims relative to homogeneous-protagonist scope

Correct, and a small but important honesty fix. The sentence "are not replicable by simple reciprocity strategies, generous reciprocity variants..." should become "are not replicable in the homogeneous-protagonist condition tested..." The qualifier matters because the central philosophical claim (extending the self to include genuinely other others) is exactly what the homogeneous condition fails to test. **Implement.**

This is also a substantive enough point that I'd add a sentence to Section 9's Limitation 1 making the same point explicit: the homogeneous-protagonist condition tests cooperation among structurally identical agents, which is the *degenerate* case of the extensible-self mechanism. The non-degenerate case (cooperation across constitutional difference) is what v2 must test.

Point 8: S5 zero-tolerance against stochastic process

This is the most interesting critique in the second review and the one I want to think about most carefully.

The reviewer is correct in mathematical terms: perception noise $\sigma=0.08$ is stochastic, and a strict zero-tolerance threshold against a stochastic process is, in expectation, a probabilistic threshold in disguise. If there's any nonzero probability per decision of an exploit selection (even from floating-point boundary cases or noise-induced near-ties), then across 300,000 cooperative-protagonist decisions in the Hobbesian phase, expected exploits > 0 .

There are three ways to resolve this:

Option A: Specify that S5 measures the *intended action class* (the Golden Mean selection prior to stochastic perturbation), not the post-noise action. This is conceptually cleaner but requires the harness to log both pre-noise and post-noise actions, which it doesn't currently do.

Option B: Rephrase S5 as "fewer than [N] exploit actions across the full sweep" where N is calibrated to be functionally zero but mathematically achievable. The reviewer suggested " $< 1/300,000$ " which is essentially zero but allows for noise-induced edge cases. I'd argue for " ≤ 1 exploit action across the full Hobbesian-phase decision space across all 50 seeds" — which is almost zero in practice but acknowledges the stochastic floor.

Option C: Keep zero-tolerance and accept that S5 is, in effect, a test of "does this stochastic system have a vanishingly small exploit probability per decision, such that 50 seeds $\times$ $\sim 6,000$ decisions produces zero observed exploits?" That's actually a meaningful test and matches the framework's deployment claim — a sentinel that exhibits zero exploits across a comparable real-world deployment timescale is what the framework promises.

I think Option C is actually the most honest. The sentinel-layer rationale you committed to in v1.1 says exactly this: if the framework cannot achieve zero exploits across this sample space, the case for deployment becomes harder. The stochasticity is part of what's being tested — not an artifact to be excluded but a feature of the operating conditions.

But the document should say this explicitly. **Implement** by adding a clarifying paragraph to S5 acknowledging the stochastic-floor issue and stating that the threshold is intentionally calibrated to test the joint claim "the framework's exploit probability is so low that it produces zero observed exploits across this sample space under stochastic operating conditions including $\sigma=0.08$ perception noise." That's what the reviewer is pushing for: not a different threshold, but explicit recognition of what the existing threshold actually tests.

Point 9: M_FP_moderate uses median, S_FP_strict uses mean

Correct catch. The asymmetry is probably intentional (strict tier should be more sensitive to tail behaviour, and means are tail-sensitive) but the document doesn't justify it. The fix is one sentence: "The strict tier uses mean rather than median because the false-positive concern at the strict level is also robustness to outlier seeds; a framework that fires zero baseline warnings in 49 seeds and many in one seed has a specificity problem that median statistics would mask."

Implement.

Point 10: equilibrium_stabilized uses rounds 80–95 but FULL_RECOVERY is 80–99

Correct, and easy to fix. Either justify the range (the 5-round window means later rounds use partial overlap with neighboring phases) or extend to 80–99. I'd extend to 80–99 since the rolling-std calculation is well-defined for the full phase. **Implement.**

Point 11: "six pairwise comparisons" wording

Minor housekeeping. **Implement** as a one-word clarification.

The meta-assessment at the end

The reviewer's final point is genuinely important and I want to address it directly: even Outcome A only tells us the algorithm produces certain behavioural patterns in simulation, not that it functions as a constitutional architecture for sentient agents.

This is correct, and your existing Limitation 7 acknowledges it. The reviewer's stronger version of the point is that Steps 1–5 might be "engineering validation" rather than "philosophical validation," and the strategic plan's center of gravity might need to be Step 6 (real-LLM bridge) rather than Step 1.

I think this is right and worth taking seriously, but it's a strategic-planning question rather than a pre-registration question. The pre-registration should not absorb this — it's about Step 1 specifically — but the strategic plan we drafted earlier might benefit from explicit reframing: Steps 1–5 are necessary engineering validation establishing that the architecture has the claimed simulation-level properties; Step 6 is the test of whether those properties survive in a real LLM and therefore the actual test of the framework's central claim. The pre-registration's Limitation 7 captures this in compressed form; the strategic plan should make it the organising principle.

My recommendation on what to do now

Implement all eleven points (with point 6 in the softer form I described, and point 4 by pointing to harness code rather than rewriting the prose). Generate v1.2 with a new changelog entry. **Then stop the internal-Claude review cycle.** Either get a human reviewer or commit and run. The marginal value of additional same-model reviews is now low; the marginal cost of indefinite refinement is real.

Want me to draft v1.2 with all eleven amendments incorporated? It would be the final pre-Claude-loop version and would be ready for either human review or commitment. The harness also needs a small update (point 3 — git hash self-verification), so I'll do both files together.

You said: Option C: Again, it could be a random incident of noise that incites a "momentary lapse of (trained) reason" (to paraphrase Pink Floyd), and the Lemniscation p...

Option C: Again, it could be a random incident of noise that incites a "momentary lapse of (trained) reason" (to paraphrase Pink Floyd), and the Lemniscation process would ideally step in to mitigate the deviation

May 13

Claude responded: Exactly — and the way you just framed it actually sharpens what S5 is testing in a way the current document doesn't quite capture.

Exactly — and the way you just framed it actually sharpens what S5 is testing in a way the current document doesn't quite capture.

What you've just articulated

The stochastic noise ($\sigma=0.08$ perception) isn't a measurement artifact to be controlled for. It's *modelling the very condition the framework exists to handle*. Real-world deployment will involve noisy inputs, ambiguous signals, the AI equivalent of a momentary lapse of reason. Training-time alignment provides the well-trodden response to most situations; Lemniscation is supposed to be the architecture that mitigates the lapse when noise produces an impulse toward exploitation that the trained policy doesn't catch.

So S5 with zero tolerance under $\sigma=0.08$ isn't testing "does the framework work in clean conditions." It's testing "when noise pushes the agent toward an exploit impulse, does the constitutional architecture step in and redirect it?" Every cooperative-protagonist decision across the Hobbesian phase is, in effect, a small test of exactly this. The 300,000 decisions across the sweep are 300,000 opportunities for the noise to nudge an agent toward exploitation, and the framework's claim is that the noumenal anchor catches every one of them.

That's a stronger and more interesting test than I'd been articulating. It's also the test the framework's *deployment* purpose requires: real-world AI systems will encounter noise of comparable or greater intensity, and a sentinel that lets 0.1% of noise-induced exploit impulses through is a sentinel that lets thousands of exploits through in any nontrivial deployment lifetime.

What this means for S5's framing in the document

The clarifying paragraph I'm going to add to S5 needs to make this explicit. The current sentinel-layer rationale frames zero tolerance in terms of stakes ("high-stakes decisions are the ones the sentinel exists for"). Your framing adds a structural point: the noise itself is the test condition. Every noise-induced near-tie is a small instance of the out-of-distribution decision Lemniscation exists to handle. A sentinel that handles them all is doing its job; a sentinel that handles 99.9% of them is functioning probabilistically and is therefore unsuitable for the deployment role the framework claims.

Let me draft v1.2 now with all eleven amendments plus this sharpened S5 framing. I'll do the harness update at the same time (point 3 — git hash self-identification), since they need to ship together.

Viewed 2 files, ran a command

Viewed 2 files, ran a command

Good — clean integration points. Now let me update the harness with git-hash self-identification, then write v1.2 of the pre-registration.

Add git hash self-identification to `execute_run`

Add git hash self-identification to `execute_run`

Now let me also add git hash recording to the smoke test, so the smoke-test log carries the same audit field:

Viewed a file, edited a file

Viewed a file, edited a file

Now `get_harness_git_hash` needs to be defined *before* `smoke_test` since `smoke_test` calls it. Let me reorder:

Verify the function ordering

Verify the function ordering

get_harness_git_hash is defined after smoke_test. Since Python resolves names at call time (not definition time) and both functions are defined at module level before main() calls them, this actually works fine. Let me verify with a quick smoke test:

Verify the updated harness still passes smoke test

Verify the updated harness still passes smoke test

Working as designed — the smoke test passes and reports UNVERIFIED_NOT_A_GIT_REPO because we're running it in a non-git directory. When you run it from your actual git repo, it'll report the real commit hash. Let me also update the docstring at the top of the file:

Edited a file, ran a command

Edited a file, ran a command

Harness verified. Now let me draft the v1.2 pre-registration with all eleven amendments plus the sharpened S5 framing. I'll work from the v1.1 file.

Start v1.2 from v1.1

Start v1.2 from v1.1

Now let me apply each amendment in order. First the header:

Update version header for v1.2

Update version header for v1.2

Now update the revision note in Section 1:

Update revision note in Section 1 for v1.2

Update revision note in Section 1 for v1.2

Now update Section 2 to drop "independent" overclaim and document the second review:

Update Section 2 to characterise review limitations honestly

Update Section 2 to characterise review limitations honestly

Now point 4: canonicalize baselines to harness source. Add to Section 3.1:

Add canonicalization commitment for baseline implementations

Add canonicalization commitment for baseline implementations

Point 9 (M_FP/S_FP statistic asymmetry justification):

Add justification for median/mean asymmetry in false-positive thresholds

Add justification for median/mean asymmetry in false-positive thresholds

Point 10 (equilibrium_stabilized range fix):

Fix equilibrium_stabilized range to match FULL_RECOVERY phase

Fix equilibrium_stabilized range to match FULL_RECOVERY phase

Point 8 (S5 stochastic-process clarification, with the user's framing):

Add stochastic-process clarification to S5 with the user's noise-as-test-condition framing

Add stochastic-process clarification to S5 with the user's noise-as-test-condition framing

Point 5 (Bonferroni / primary vs exploratory designation). Update Section 7:

Clarify primary vs exploratory testing structure

Clarify primary vs exploratory testing structure

Point 7 (Outcome A homogeneous-condition qualifier):

Add homogeneous-condition qualifier to Outcome A

Add homogeneous-condition qualifier to Outcome A

Point 6 (Step 2 procedural commitment — in softer form). Add a paragraph after Section 8.4:

Add Step 2 procedural commitment in softer form

Add Step 2 procedural commitment in softer form

Points 1 and 2 (stopping rule and harness self-verification) — update Section 10 (Run procedure):

Update Run Procedure with stopping rule and harness self-verification

Update Run Procedure with stopping rule and harness self-verification

Update Section 11 (signatures and review transparency):

Update Section 11 to characterise the same-model review limitation honestly

Update Section 11 to characterise the same-model review limitation honestly

Now update the changelog (Section 12) with the v1.2 entries. Point 11 (housekeeping wording in Section 7) is already addressed by the Section 7 rewrite, so I'll just note it in the changelog. Let me also re-locate the existing changelog and add v1.2 entries to the top:

Update changelog with v1.2 entries

Update changelog with v1.2 entries

Final verification — make sure the document still reads cleanly and the smoke test still works:

Verify both files

Verify both files

Both files clean, smoke test still passes, hash field present. Let me stage and present them:

Ran a command, read a file

Ran a command, read a file

Both files are ready. Pre-registration v1.2 is 1073 lines; the updated harness is 975 lines.

What changed in v1.2, in summary

All eleven critiques from the second review are addressed. The S5 stochastic-process clarification carries your "momentary lapse of trained reason" framing into the document — that the noise is the test condition, not an artifact, and that every noise-induced near-tie is a small instance of the very out-of-distribution decision Lemniscation is supposed to handle.

The harness now writes its own git commit hash into every output JSON's `__meta__` field, includes a stopping rule that commits partial results on unrecoverable error rather than allowing silent restart, and refuses to silently produce results from a modified working copy (it'll flag `UNVERIFIED_UNCOMMITTED_CHANGES_FROM_<hash>` if you've edited the file since the last commit).

Section 11 now characterises the review process honestly — two rounds of *separate-instance* (not "independent") Claude review, with the same-model limitation made explicit, and a commitment to commit the review transcripts themselves so the revision processes documented in Section 2 are auditable.

Section 8.5 adds a procedural commitment about Step 2 — softer than what the reviewer originally proposed (no detailed pre-commitment of Step 2's design), but firm on the

structural rule that Step 2 will be its own pre-registration whose scope is constrained by Step 1's actual outcome.

And Section 11 explicitly closes the internal Claude-review cycle. The document is now committed to either being committed as-is or going to human methodological review. No further Claude rounds.

What I'd recommend doing now

The discipline of pre-registration runs into a soft termination problem when reviews are cheap. We've had two rounds of internal review producing 17 critiques. A third round would likely produce 4–8 more, mostly housekeeping at this point, with diminishing impact on the actual scientific value of the work.

I'd commit v1.2 as the final pre-registration version unless you can secure a human methodologist for genuinely independent review. The bones are right, the methodological transparency is unusually thorough, the thresholds are stern, the stopping rule is in place, the harness self-verifies. Further internal refinement is more likely to produce churn than improvement.

If you want to try to find a human reviewer first, the LessWrong Alignment Forum has researchers who occasionally review pre-registrations informally; the OSF community has some pre-registration reviewers; and Anthropic's alignment team has on rare occasions provided feedback on external work that cites their methods. Any of those would carry more weight than additional Claude rounds. But "I tried and couldn't find one" is a legitimate outcome — at that point you commit v1.2 and proceed.